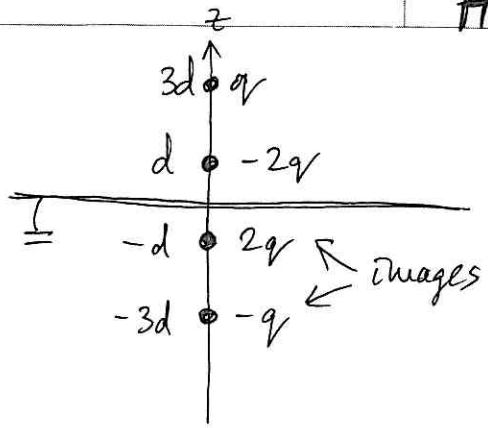
P3.6



$$\vec{F} = \frac{-\hat{z}}{4\pi\varepsilon_0}\left(\frac{2q^2}{(2d)^2} + \frac{q^2}{(6d)^2} - \frac{2q^2}{(4d)^2}\right)$$

$$= -\hat{z}\,\frac{q^2}{4\pi\varepsilon_0 d^2}\underbrace{\left(\frac{1}{2} + \frac{1}{36} - \frac{1}{8}\right)}_{\frac{29}{72}}$$

P3.7

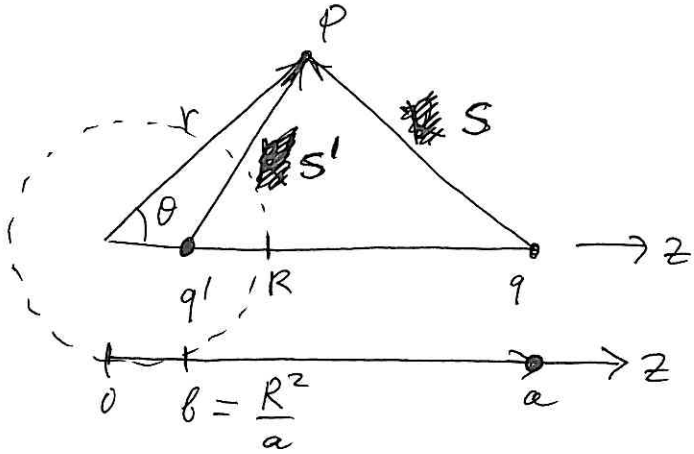


Image charge

$$q' = -\frac{R}{a}q$$

$$b = \frac{R^2}{a}$$

(a) $\vec{s} = \vec{r} - \vec{a} \Rightarrow s^2 = r^2 + a^2 - 2\vec{r}\vec{a} = r^2 + a^2 - 2ra\cos\theta$

$\vec{s}\,' = \vec{r} - \vec{b} \Rightarrow (s')^2 = r^2 + b^2 - 2rb\cos\theta$

$$V(r) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{s} + \frac{q'}{s'}\right) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{r^2+a^2-2ra\cos\theta}}\right.$$

$$\left. - \frac{Rq}{a\sqrt{r^2+b^2-2rb\cos\theta}}\right) = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{\sqrt{r^2+a^2-2ra\cos\theta}} - \frac{q}{\sqrt{R^2+\left(\frac{ar}{R}\right)^2-2ra\cos\theta}}\right);$$

But $\frac{R}{a}\,\frac{1}{\sqrt{r^2+b^2-2rb\cos\theta}} = \frac{1}{\sqrt{\frac{a^2r^2}{R^2} + \frac{a^2}{R^2}\left(\frac{R^2}{a}\right)^2 - 2\frac{a^2}{R^2}r\left(\frac{R^2}{a}\right)\cos\theta}}$

$$= \frac{1}{\left[R^2 + \left(\frac{ar}{R}\right)^2 - 2ra\cos\theta\right]^{1/2}}.$$ Thus $V(r=R) = 0$.

(b) Boundary conditions for conducting sphere imply

$$E_{\perp,\text{out}} - \underset{\substack{\| \\ 0}}{E_{\perp,\text{inside}}} = \frac{\sigma}{\varepsilon_0}$$

$$E_{\perp,\text{out}} = -\frac{\partial V}{\partial r}\Big|_{r=R} = \frac{-q}{4\pi\varepsilon_0}\left\{\frac{-\frac{1}{2}(2r-2a\cos\theta)}{(r^2+a^2-2ra\cos\theta)^{3/2}}\right.$$

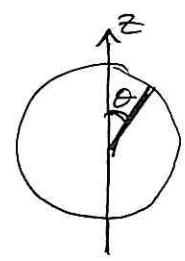
$$\left. - \frac{\left(-\frac{1}{2}\right)\left(2r\frac{a^2}{R^2}-2a\cos\theta\right)}{\left(R^2+\left(\frac{ar}{R}\right)^2-2ra\cos\theta\right)^{3/2}}\right\}_{r=R} = \frac{q}{4\pi\varepsilon_0}\left\{\frac{R-a\cos\theta}{(R^2+a^2-2Ra\cos\theta)^{3/2}}\right.$$

$$\left. - \frac{\frac{a^2}{R}-a\cos\theta}{(R^2+a^2-2Ra\cos\theta)^{3/2}}\right\} = \frac{1}{4\pi\varepsilon_0}\frac{\left(R-\frac{a^2}{R}\right)q}{(R^2+a^2-2Ra\cos\theta)^{3/2}}$$

$$\Rightarrow \sigma = \frac{q}{4\pi}\frac{R\left(1-\frac{a^2}{R^2}\right)}{R^3\left(1+\frac{a^2}{R^2}-\frac{2a}{R}\cos\theta\right)^{3/2}}$$

Integrate over the surface now:

$$Q = \int_0^{\pi}\sigma\, R^2\int_0^{2\pi}d\varphi\,\sin\theta\, d\theta.$$



Denote $\frac{a}{R} = c$. Note that $c>1$:

$$Q = q\,\frac{2\pi R^2}{4\pi}\,\frac{(1-c^2)}{R^2}\underbrace{\int_{-1}^{1} d(\cos\theta)\,\frac{1}{(1+c^2-2c\cos\theta)^{3/2}}}_{\frac{1}{c}(1+c^2-2c\cos\theta)^{-1/2}\Big|_{-1}^{1}} =$$

$$= \frac{q}{2}(1-c^2)\left(\frac{1}{c}\frac{1}{\sqrt{1+c^2-2c}} - \frac{1}{c\sqrt{1+c^2+2c}}\right) =$$

$$= q\,\frac{1-c^2}{2c}\left(\frac{1}{|1-c|}-\frac{1}{1+c}\right) = q\,\frac{1-c^2}{2c}\,\frac{1+c-(c-1)}{c^2-1} = \frac{-q}{c} = -\frac{R}{a}q.$$

Of course, as it should be $Q = q' = -\frac{R}{a}q$ the ~~[illegible]~~ image charge.

(c) Electrostatic energy is best calculated via work required to bring the charge q from $r=\infty$ to $r=a$:

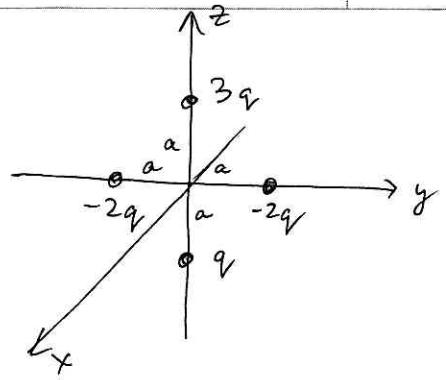
$$W = \int_{\infty}^{a} \vec{F}\cdot d\vec{l} = \frac{1}{4\pi\varepsilon_0}\int_{\infty}^{a} \frac{q^2 R\, r}{(r^2-R^2)^2}\, dr =$$

$(X=r^2)$

$$= \frac{q^2 R}{8\pi\varepsilon_0}\int_{\infty}^{a^2} \frac{dX}{(X-R^2)^2} = -\frac{q^2 R}{8\pi\varepsilon_0}\,\frac{1}{X-R^2}\Big|_{\infty}^{a^2} =$$

$$= -\frac{1}{8\pi\varepsilon_0}\,\frac{q^2 R}{a^2-R^2}\,.$$

P3.27



$$\vec{r} = (x, y, z)$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left(\frac{-2q}{|\vec{r}-a\hat{x}|} + \frac{-2q}{|\vec{r}+a\hat{x}|} + \frac{3q}{|\vec{r}-a\hat{y}|} + \frac{q}{|\vec{r}+a\hat{y}|}\right)$$

$$|\vec{r}-a\hat{x}|^2 = (\vec{r}-a\hat{x})^2 = r^2+a^2-2\,\vec{r}\cdot a\hat{x} =$$
$$= r^2+a^2-2ax$$

$$|\vec{r}-a\hat{y}| = r^2+a^2-2ay$$

$$\frac{1}{\sqrt{r^2+a^2-2ax}} = \frac{1}{r\left(1+\frac{a^2}{r^2}-\frac{2ax}{r^2}\right)^{1/2}} = \frac{1}{r}\left\{1-\frac{1}{2}\left(\frac{a^2}{r^2}-\frac{2ax}{r^2}\right)+\right.$$

$$\left.+\frac{3}{8}\left(\frac{a^2}{r^2}-\frac{2ax}{r^2}\right)^2+\dots\right\} = \frac{1}{r}\left\{1+\frac{ax}{r^2}-\frac{a^2}{2r^2}+\frac{3a^2x^2}{2r^4}+O\left(\frac{1}{r^3}\right)\right\}$$

$$\Rightarrow V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left\{-\frac{2q}{r}\left(2-\frac{a^2}{r^2}+\frac{3a^2x^2}{r^4}\right)+\right.$$

$$\left.+\frac{q}{r}\left[3\left(1+\frac{ay}{r^2}-\frac{a^2}{2r^2}+\frac{3a^2y^2}{2r^4}\right)+1-\frac{ay}{r^2}-\frac{a^2}{2r^2}+\frac{3a^2y^2}{2r^4}\right]\right\}$$

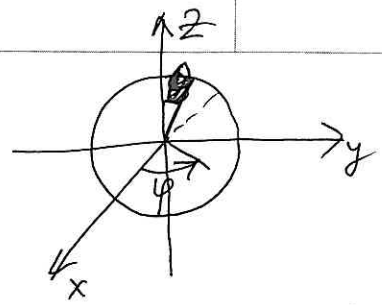
$$= \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{r}\underbrace{\left(-4+3+1\right)}_{0} + \frac{1}{r^3}\underbrace{\left(2a^2-\frac{3}{2}a^2-\frac{a^2}{2}+3ay-ay\right)}_{2ay}+\dots\right\}$$

total charge $Q=0$

$$= \frac{1}{4\pi\varepsilon_0}\frac{2aqy}{r^3} = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\hat{y}}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\hat{r}}{r^2}$$

where dipole moment $\vec{p} = 2aq\hat{y}$

P3.28



$\sigma = k\cos\theta$

(a) $\vec{p} = \int \vec{r}\,\rho(\vec{r})\,dV = \overset{\text{surface integral}}{\int} \left(R\sin\theta\left(\cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}\right) + R\cos\theta\,\hat{z}\right) \cdot$

$\cdot\; k\cos\theta \cdot R^2 \sin\theta\, d\theta\, d\varphi =$

$= k R^3 \hat{z} \int_{-1}^{1} \cos^2\theta\, d(\cos\theta) \int_0^{2\pi} d\varphi = \frac{4\pi}{3} k R^3 \hat{z}.$

$\int_0^{2\pi} d\varphi \cos\varphi = \int_0^{2\pi} d\varphi \sin\varphi = 0$; $\int_{-1}^{1} d(\cos\theta)\cos^2\theta = \frac{1}{3}\cos^3\theta\Big|_{-1}^{1} = \frac{2}{3}.$

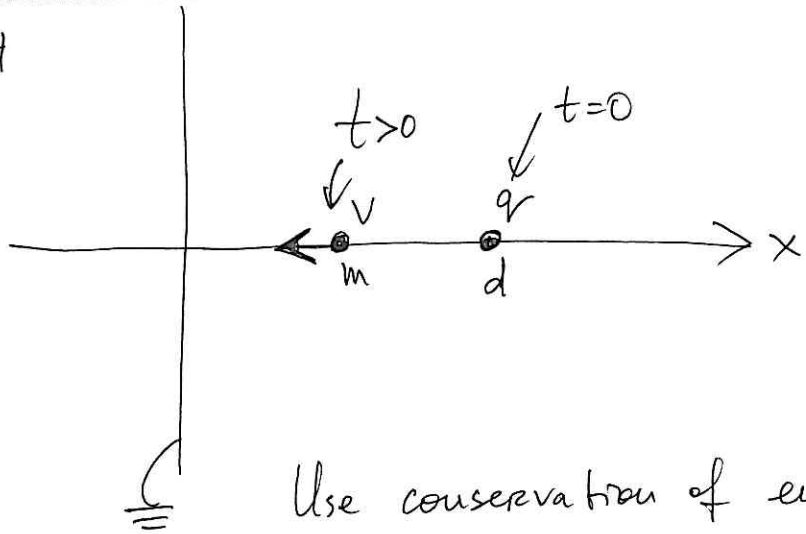
$\Rightarrow$ dipole moment is oriented along $\hat{z}$-axis.

(b) $V_{dip}(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{\vec{p}\cdot\hat{r}}{r^2} = \frac{1}{4\pi\varepsilon_0} \frac{\left(\frac{4\pi}{3} k R^3 \hat{z}\right) \cdot \left(\sin\theta\cos\varphi\,\hat{x} +\right.}{r^2}$

$\left. + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}\right) = \frac{k R^3}{3\varepsilon_0} \frac{\cos\theta}{r^2}$

which coincides with eq.(3.87) exactly.

P3.34



Use conservation of energy:

$$E_{initial} = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4d}$$

$$E(t) = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{4x} + \frac{1}{2}mv^2 \quad , \text{for } x<d.$$

$$\vec{v} = \frac{dx}{dt}\hat{x}$$

$$\Rightarrow \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 = \frac{q^2}{16\pi\varepsilon_0}\left(\frac{1}{x}-\frac{1}{d}\right)$$

choose negative sign:

$$\frac{dx}{dt} = -\sqrt{\frac{q^2}{8\pi\varepsilon_0 m}}\left(\frac{1}{x}-\frac{1}{d}\right)^{1/2}$$

$$\int_d^0 \frac{dx}{\sqrt{\frac{1}{x}-\frac{1}{d}}} = -\frac{q}{\sqrt{8\pi\varepsilon_0 m}}\int_0^t dt = -\frac{qt}{\sqrt{8\pi\varepsilon_0 m}}$$

$x = ds$

$$\int_1^0 \frac{d\,ds}{\frac{1}{\sqrt{d}}\sqrt{\frac{1}{s}-1}} = d^{3/2}\int_1^0 \frac{\sqrt{s}\,ds}{\sqrt{1-s}}$$

let now $s = \cos^2\alpha \Rightarrow I = +\int_0^{\pi/2}\frac{\cos\alpha\, d(\cos^2\alpha)}{\sqrt{1-\cos^2\alpha}} =$

$$= -\int_0^{\pi/2}\frac{\cos\alpha}{\sin\alpha}\,2\cos\alpha\sin\alpha\,d\alpha = -2\int_0^{\pi/2}\cos^2\alpha\,d\alpha = -2\left(\frac{\pi}{4}\right) = -\frac{\pi}{2}$$

Thus $\left(-\frac{\pi}{2}\right) d^{3/2} = -\frac{q\,t}{\sqrt{8\pi\varepsilon_0 m}}$

$$t = \frac{\pi}{2}\frac{\sqrt{8\pi\varepsilon_0 m}}{q}\, d^{3/2} = \frac{\pi d^{3/2}}{q}\sqrt{2\pi\varepsilon_0 m}$$